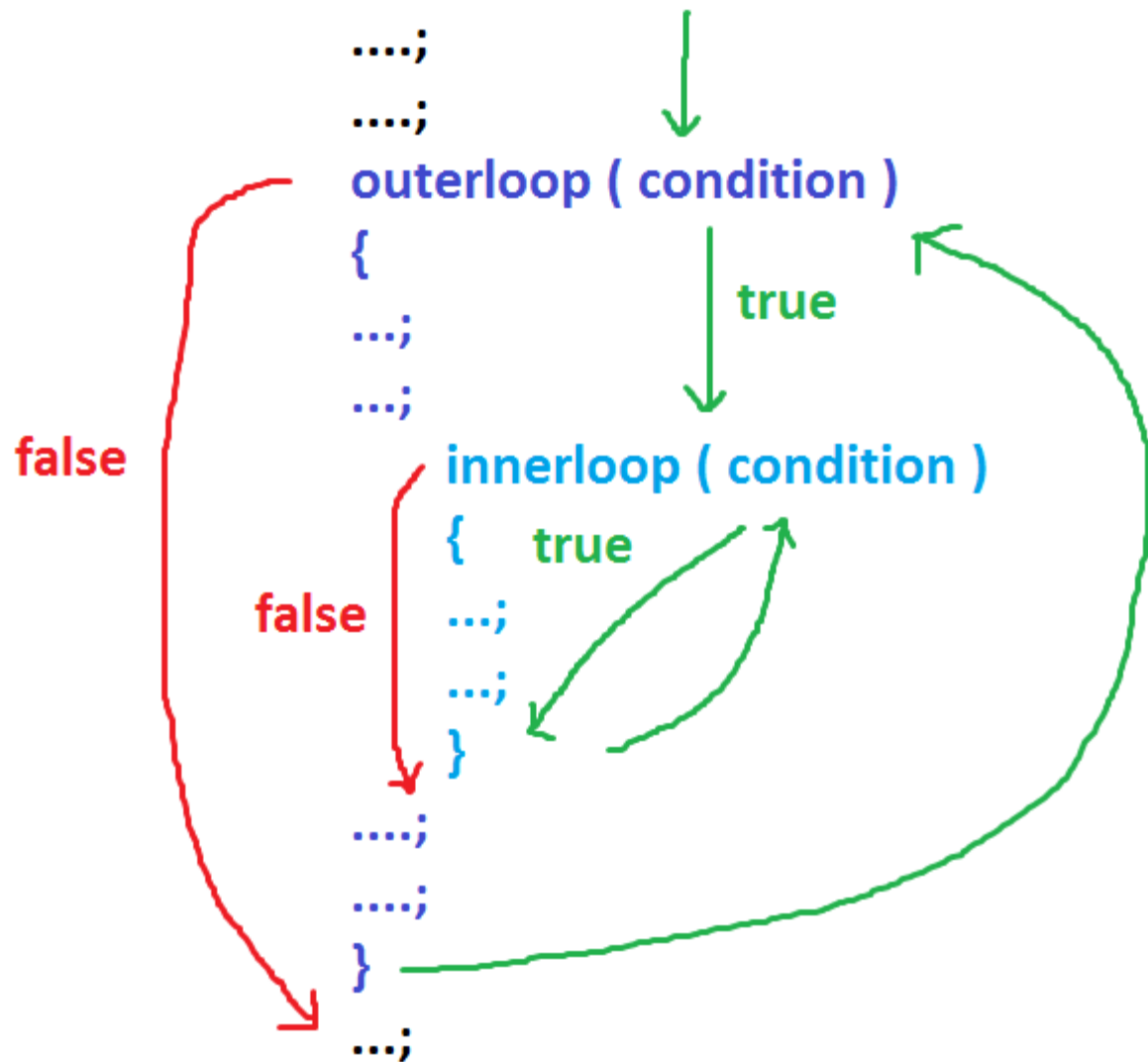
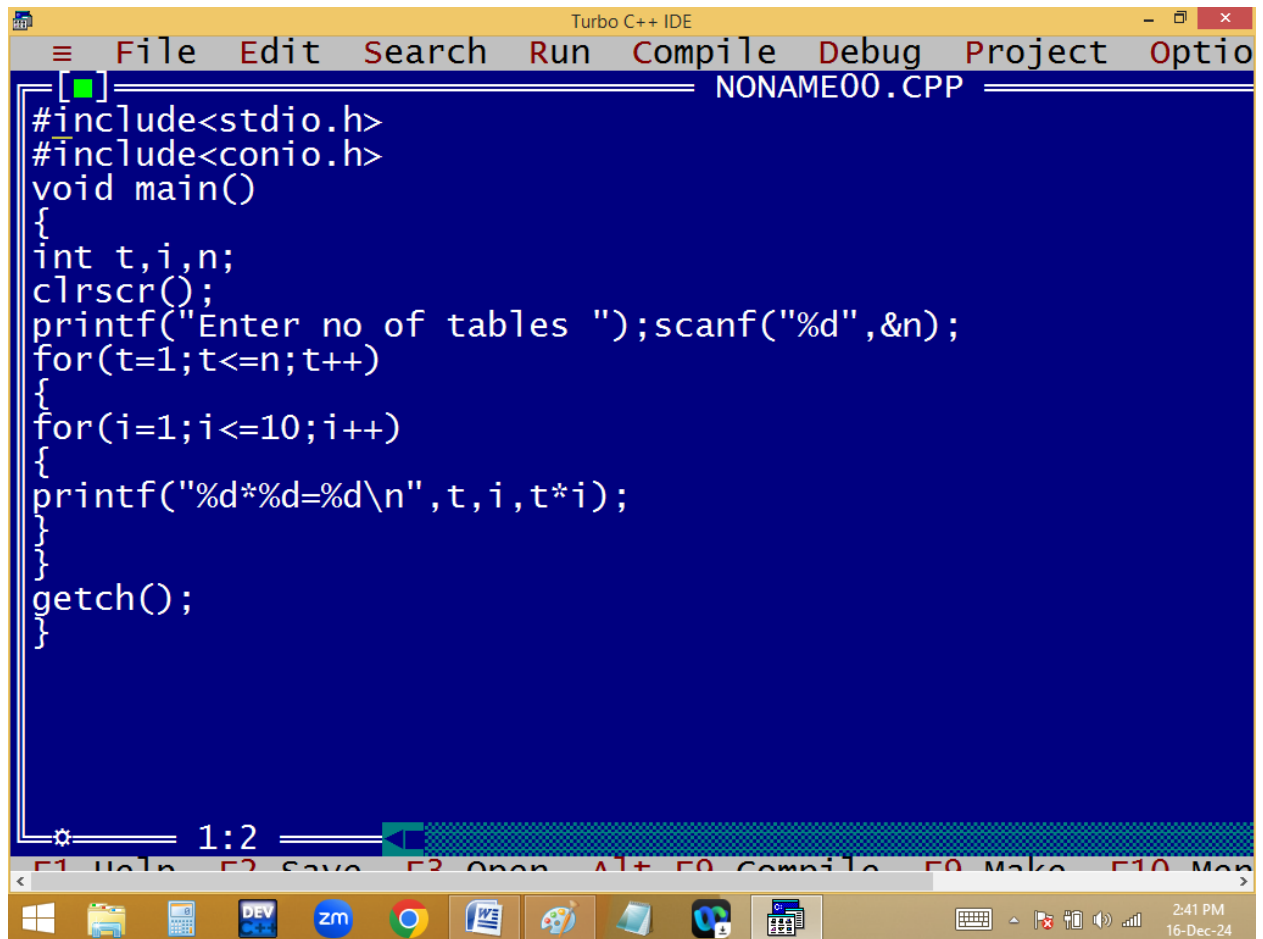


**Nested loops:** A loop within another loop is called nested loop.



**Printing 1 to n no of tables:**

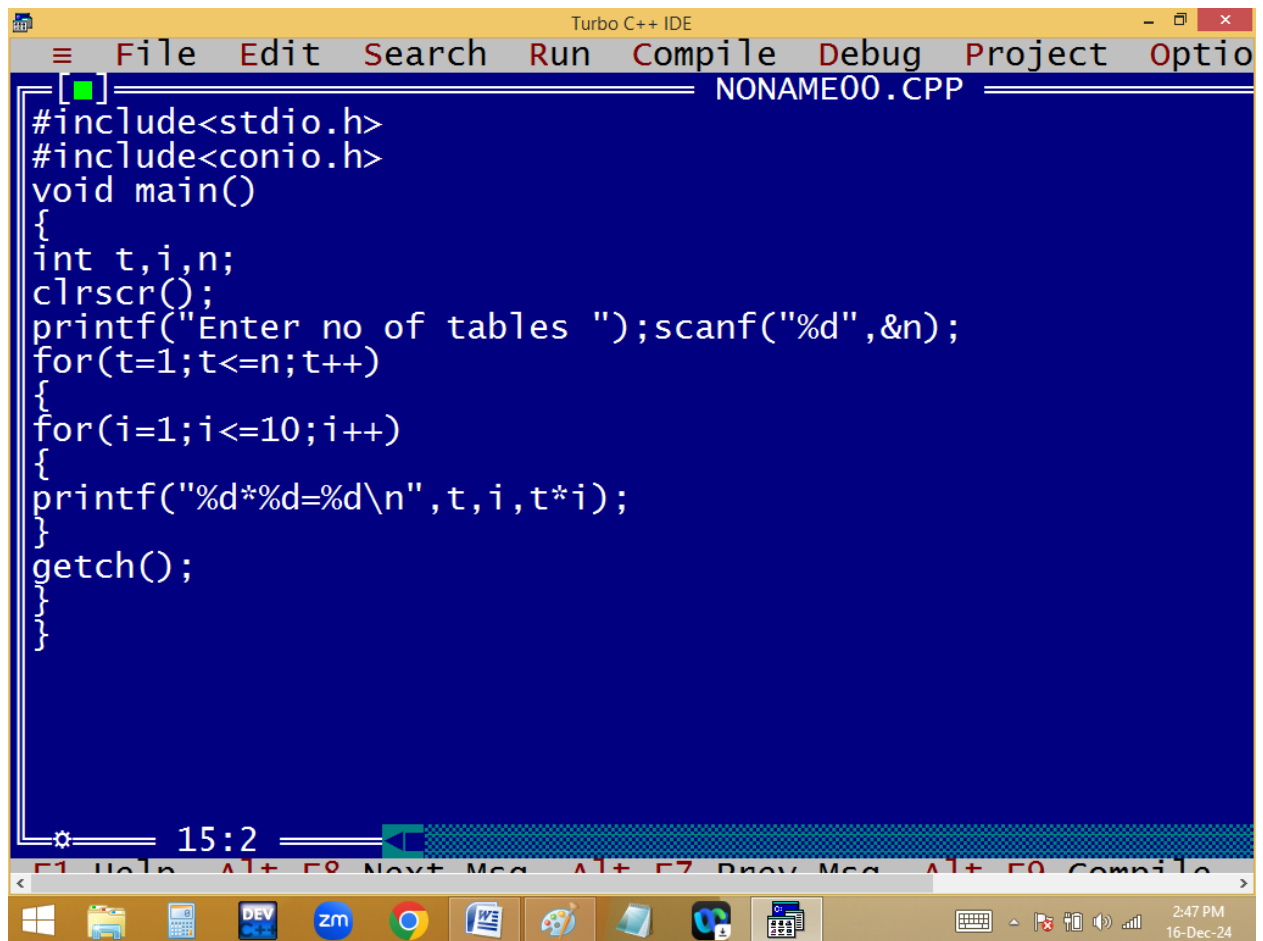


The image shows a screenshot of the Turbo C++ IDE. The title bar at the top reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "NONAME00.CPP" is displayed in the top right corner of the editor window. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables ");scanf("%d",&n);
for(t=1;t<=n;t++)
{
for(i=1;i<=10;i++)
{
printf("%d*%d=%d\n",t,i,t*i);
}
}
getch();
}
```

The status bar at the bottom left shows the cursor position "1:2". The Windows taskbar is visible at the very bottom, showing icons for various applications and the system clock indicating "2:41 PM 16-Dec-24".

```
Turbo C++ IDE
Enter no of tables 2
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20
_
```

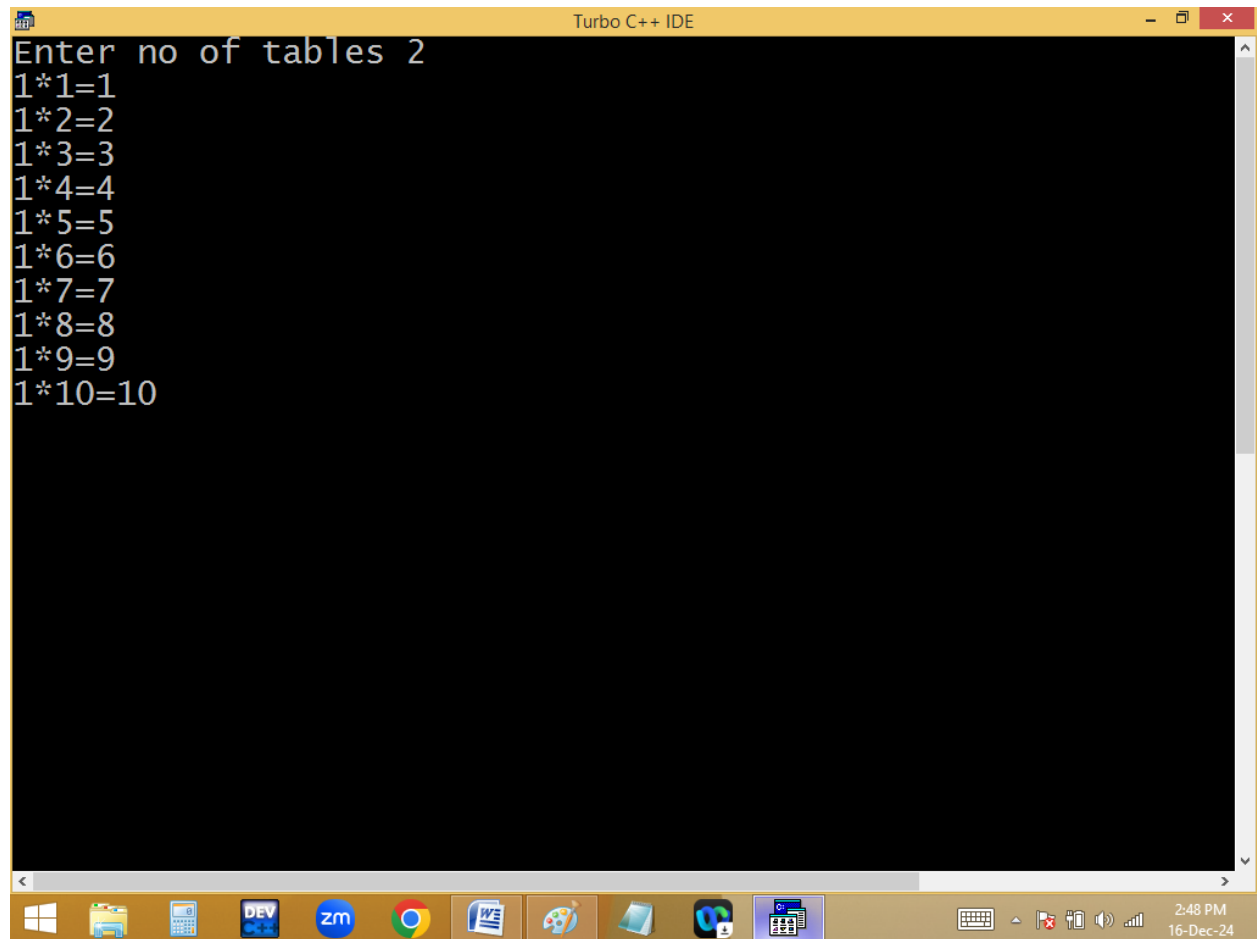


The image shows a screenshot of the Turbo C++ IDE. The title bar at the top reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "NONAME00.CPP" is displayed in the top right corner of the editor window. The code in the editor is as follows:

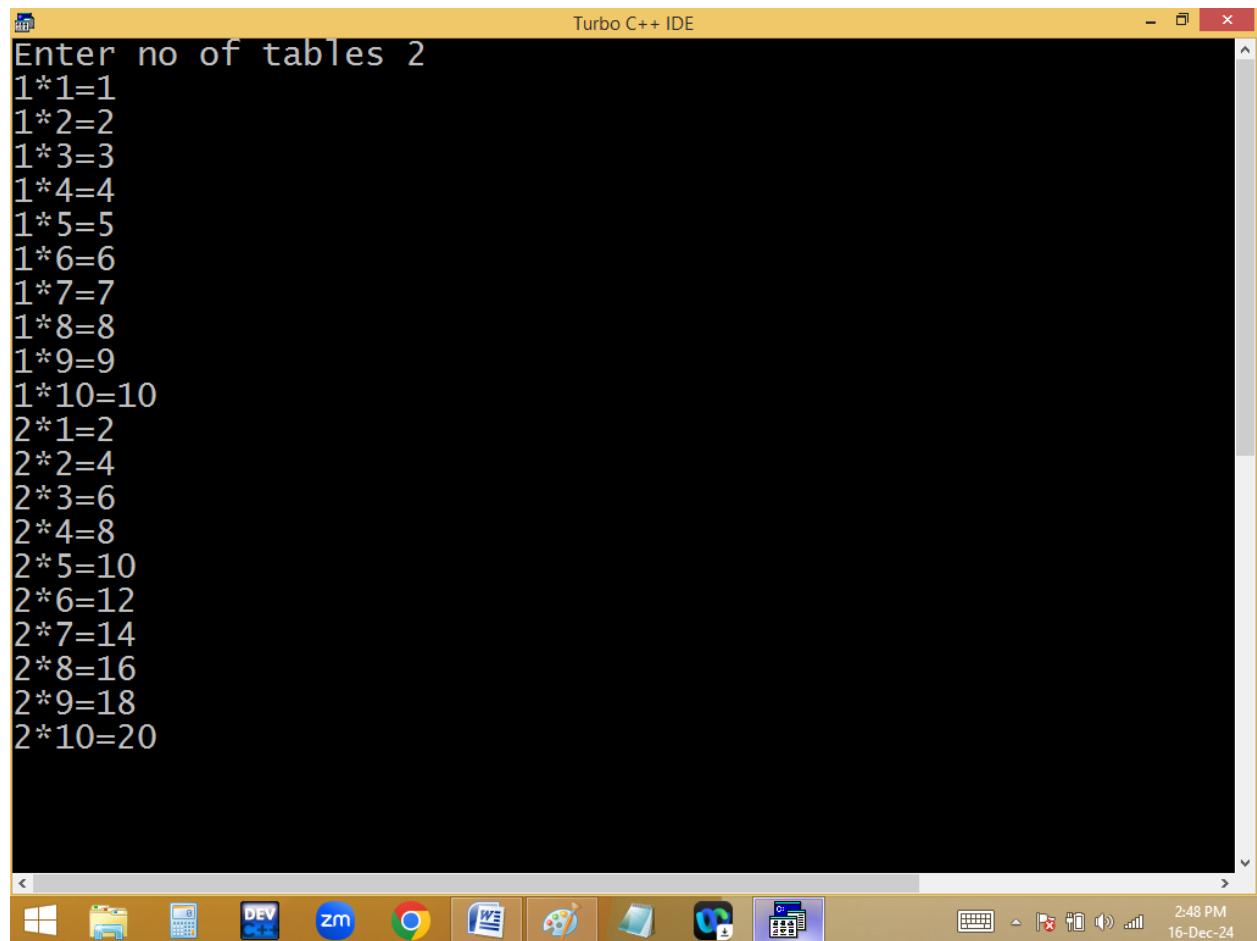
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)
        {
            printf("%d*%d=%d\n",t,i,t*i);
        }
        getch();
    }
}
```

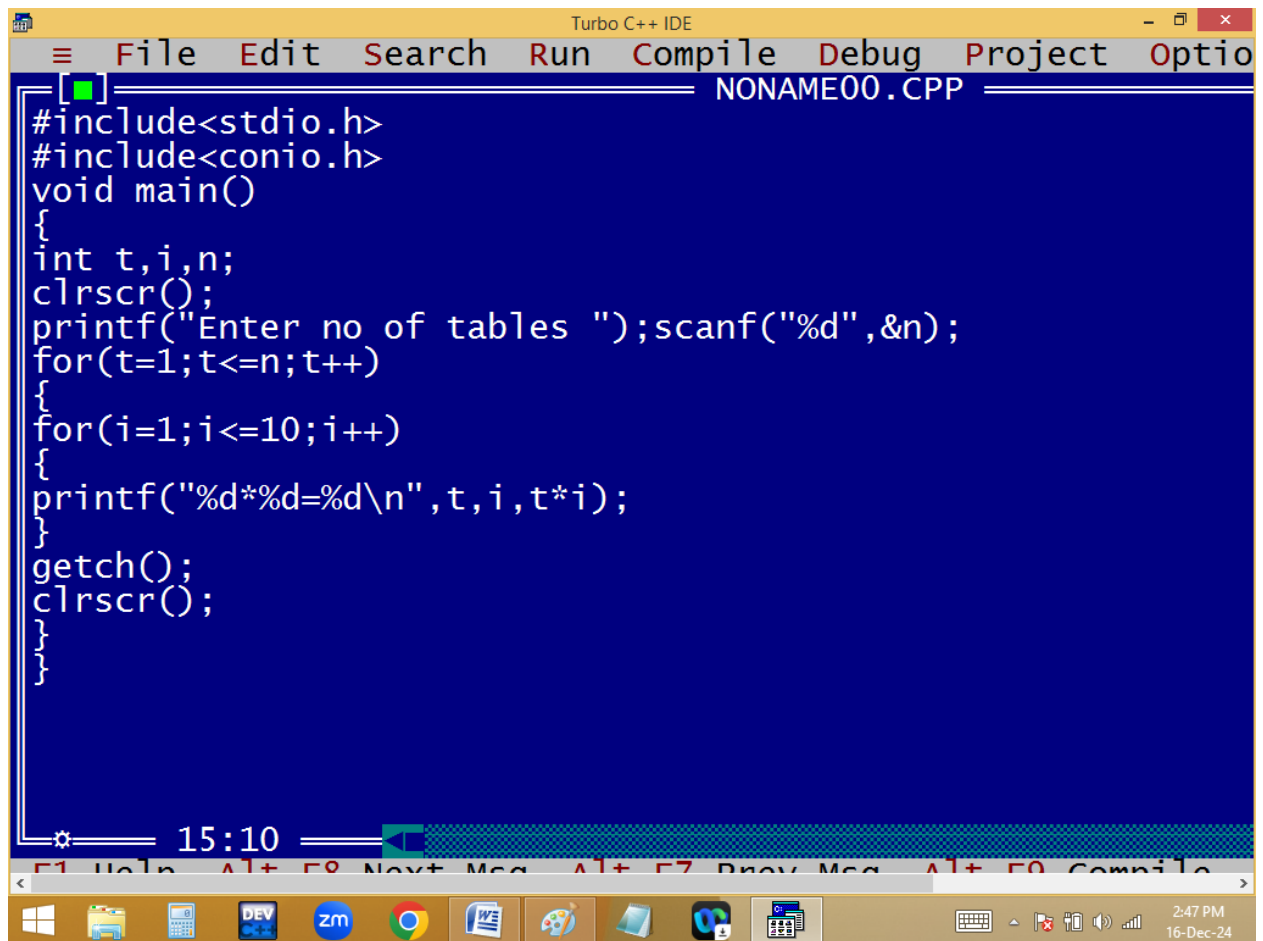
At the bottom of the IDE window, there is a status bar showing the time "15:2" and a cursor icon. Below the IDE window, the Windows taskbar is visible, showing icons for various applications including a folder, calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a game controller. The system tray on the right shows the time "2:47 PM" and the date "16-Dec-24".

```
Turbo C++ IDE
Enter no of tables 2
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
```

The image shows a screenshot of the Turbo C++ IDE window. The title bar at the top reads "Turbo C++ IDE". The main text area contains the output of a program that prints multiplication tables for the number 1, from 1 to 10. The text is as follows: "Enter no of tables 2", "1\*1=1", "1\*2=2", "1\*3=3", "1\*4=4", "1\*5=5", "1\*6=6", "1\*7=7", "1\*8=8", "1\*9=9", and "1\*10=10". The IDE has a yellow title bar and a black text area. At the bottom, there is a Windows taskbar with various application icons including Windows Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and the Turbo C++ IDE itself. The system clock in the bottom right corner shows "2:48 PM" and "16-Dec-24".

```
Turbo C++ IDE
Enter no of tables 2
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20
```

The image shows a screenshot of the Turbo C++ IDE window. The title bar at the top reads "Turbo C++ IDE". The main text area is black with white text. It starts with the prompt "Enter no of tables 2". Below this, it displays two multiplication tables. The first table is for the number 1, showing products from 1\*1 to 1\*10. The second table is for the number 2, showing products from 2\*1 to 2\*10. At the bottom of the window is a Windows taskbar with various application icons including Windows Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder icon. System tray icons for keyboard, volume, and network are also visible, along with the date and time "2:48 PM 16-Dec-24".

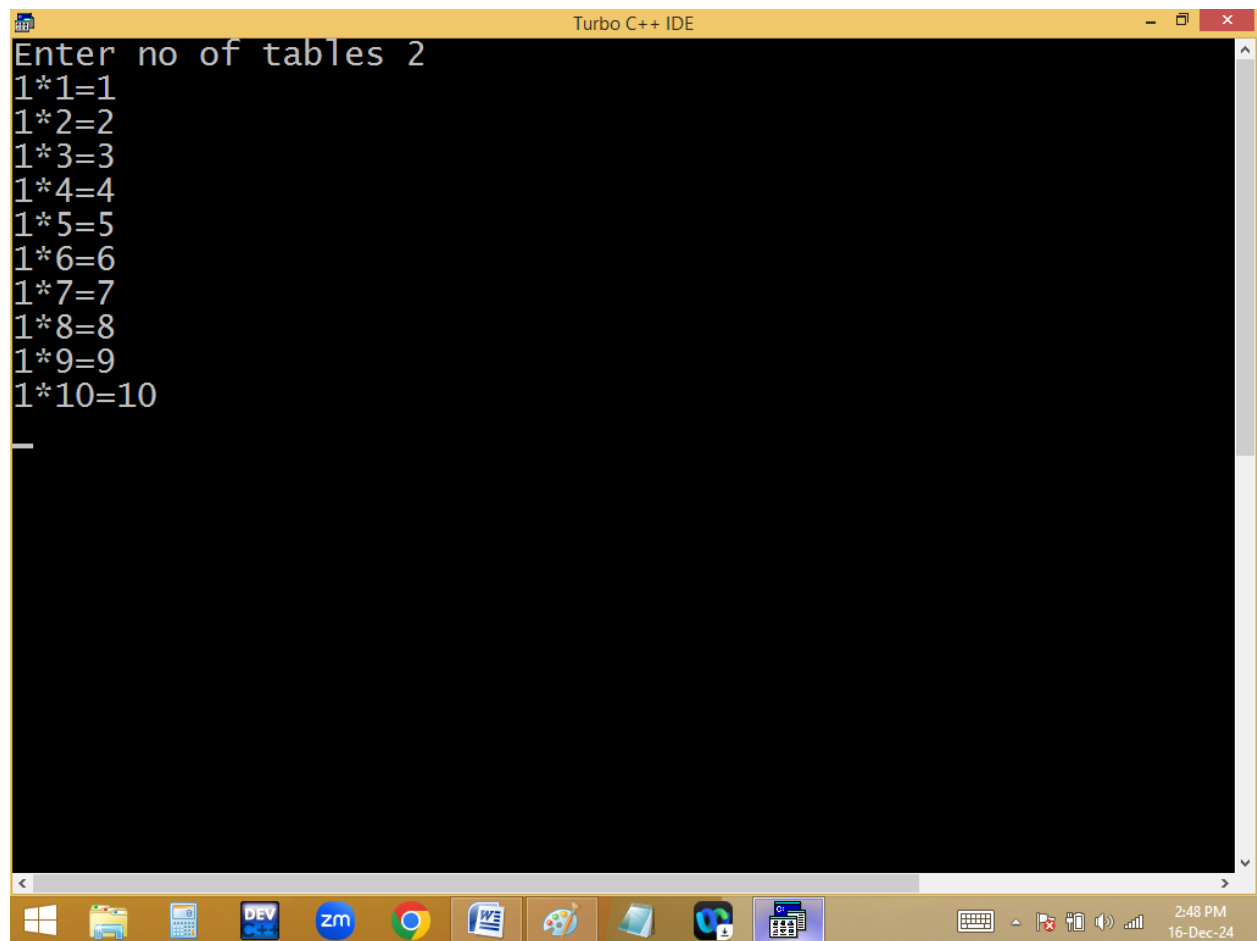


The image shows a screenshot of the Turbo C++ IDE. The title bar at the top reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "NONAME00.CPP" is displayed in the top right corner of the editor window. The code in the editor is as follows:

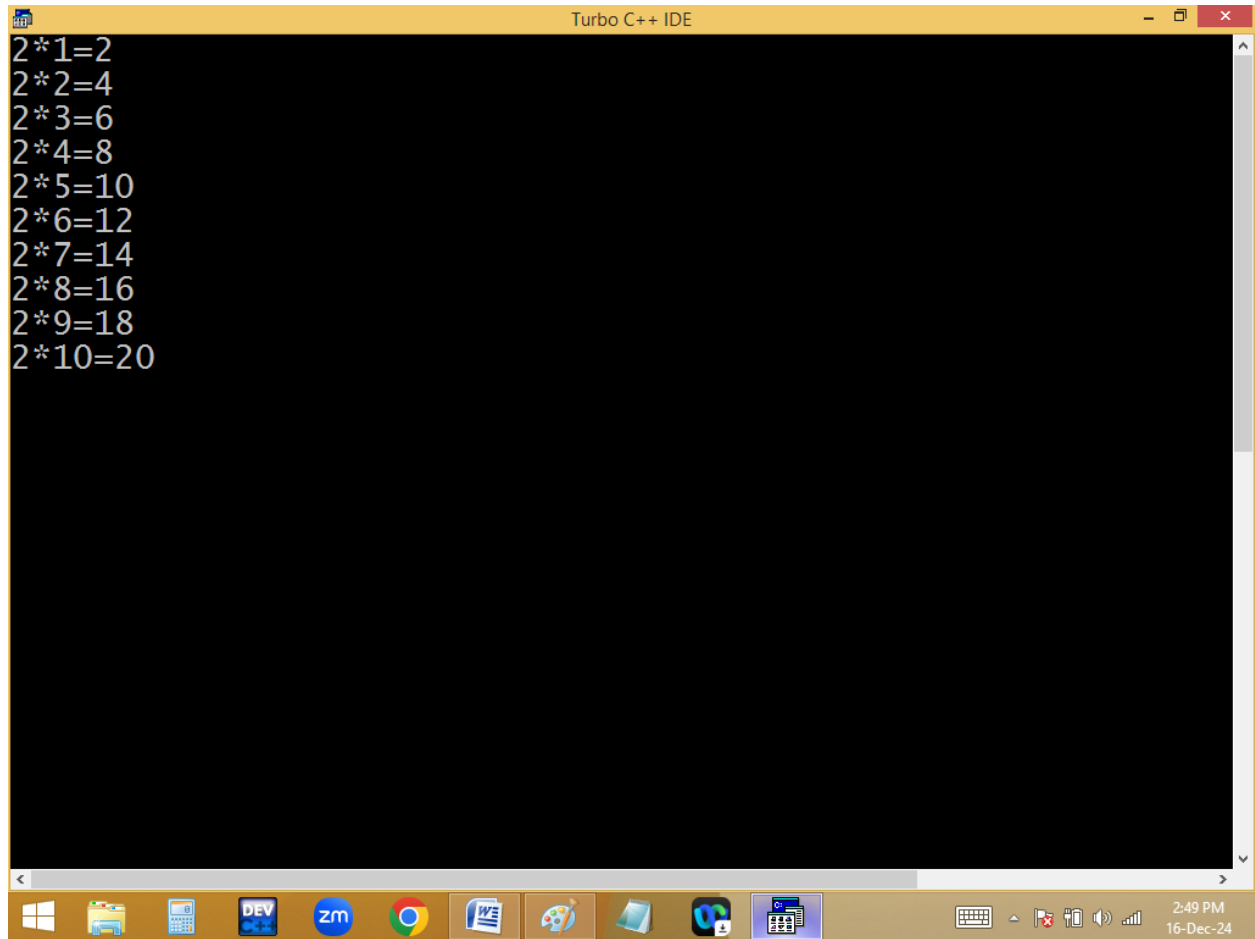
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)
        {
            printf("%d*%d=%d\n",t,i,t*i);
        }
        getch();
        clrscr();
    }
}
```

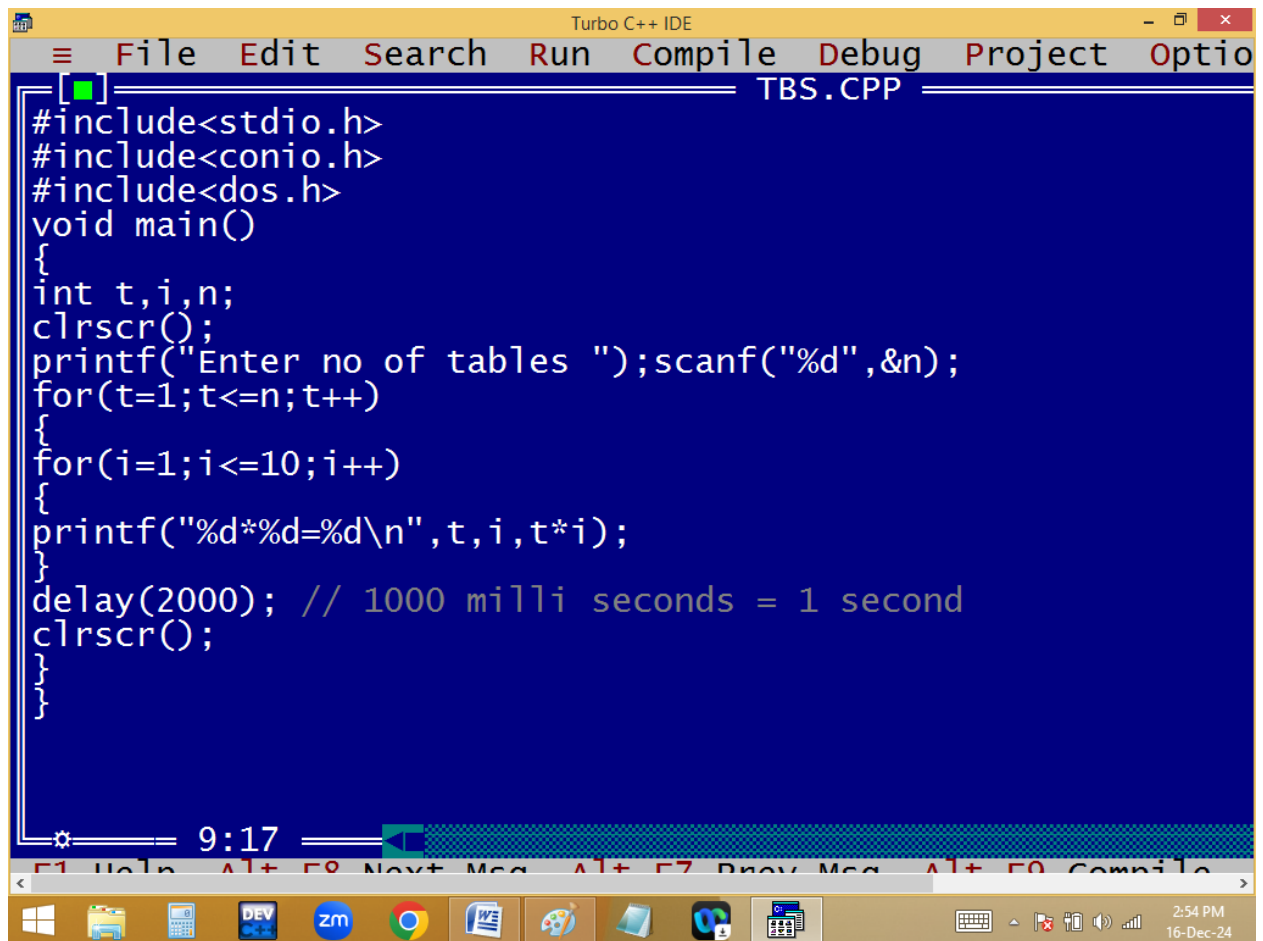
At the bottom of the IDE window, there is a status bar showing the time "15:10" and a keyboard icon. Below the IDE window, the Windows taskbar is visible, showing icons for various applications including a folder, calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a game controller. The system tray on the right shows the time "2:47 PM" and the date "16-Dec-24".

```
Turbo C++ IDE
Enter no of tables 2
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
```





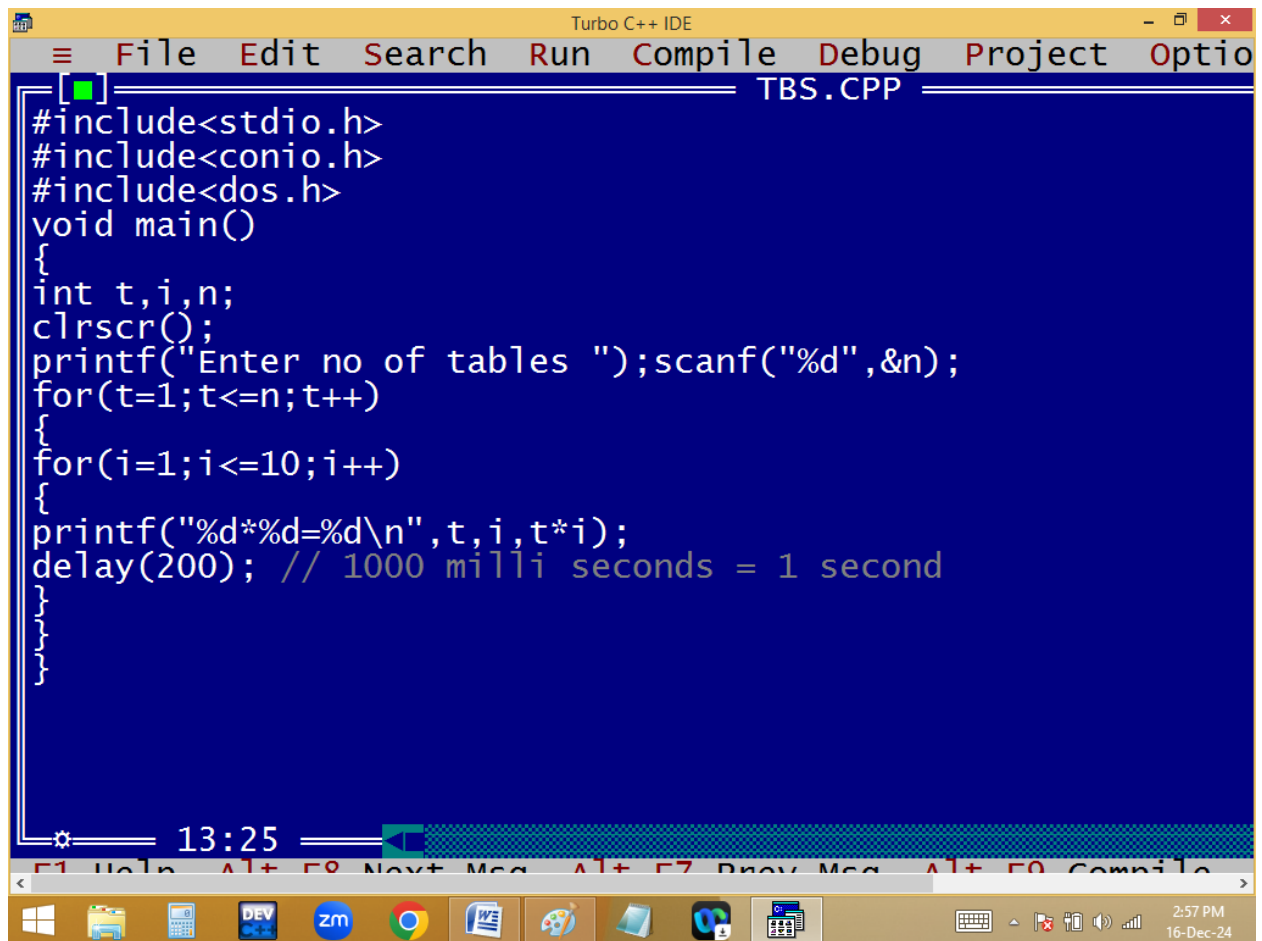




The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<dos.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)
        {
            printf("%d*%d=%d\n",t,i,t*i);
        }
        delay(2000); // 1000 milli seconds = 1 second
        clrscr();
    }
}
```

The status bar at the bottom shows the time "9:17" and a keyboard icon. The Windows taskbar at the very bottom includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time "2:54 PM" and the date "16-Dec-24".

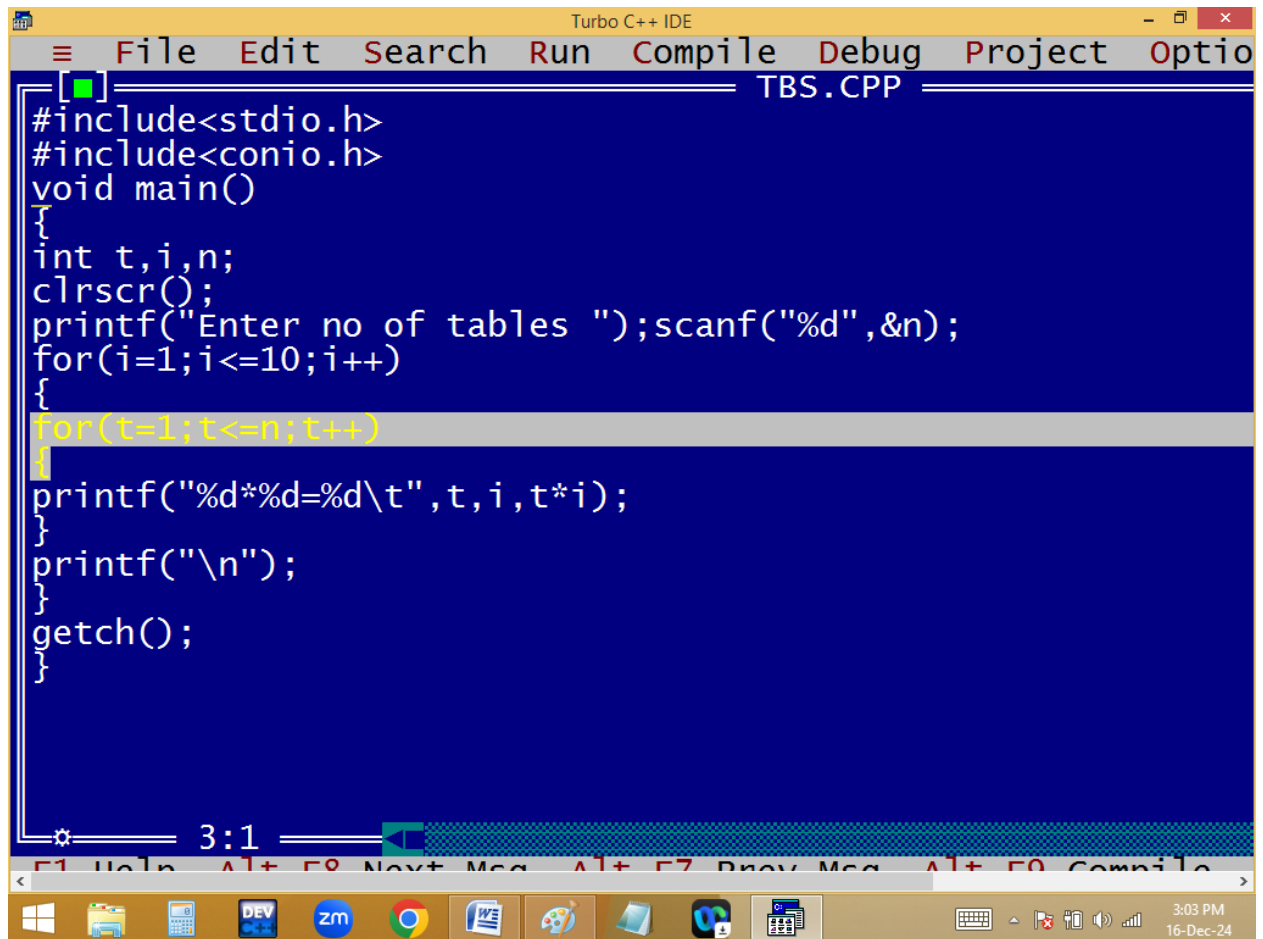


The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<dos.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)
        {
            printf("%d*%d=%d\n",t,i,t*i);
            delay(200); // 1000 milli seconds = 1 second
        }
    }
}
```

At the bottom of the IDE, there is a status bar showing the time "13:25" and a keyboard icon. Below the IDE window, the Windows taskbar is visible with icons for Windows, File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time "2:57 PM" and the date "16-Dec-24".

Side by side:

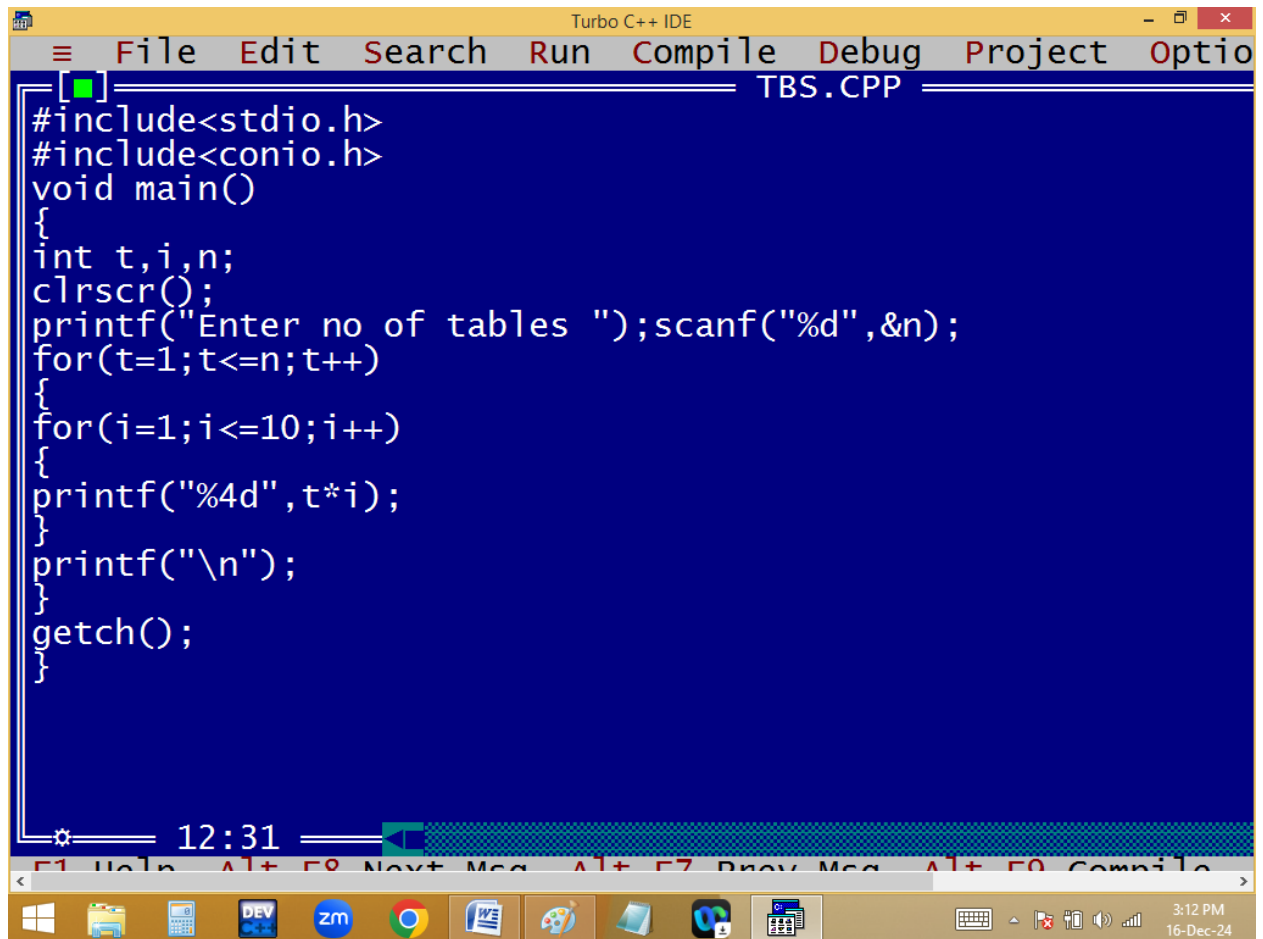


The image shows a screenshot of the Turbo C++ IDE. The title bar at the top reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right corner of the editor window. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(i=1;i<=10;i++)
    {
        for(t=1;t<=n;t++)
        {
            printf("%d*%d=%d\t",t,i,t*i);
        }
        printf("\n");
    }
    getch();
}
```

The status bar at the bottom left shows the cursor position "3:1". The Windows taskbar at the bottom contains icons for various applications, including a file explorer, calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner indicates the time is 3:03 PM on 16-Dec-24.

```
Turbo C++ IDE
Enter no of tables 3
1*1=1    2*1=2    3*1=3
1*2=2    2*2=4    3*2=6
1*3=3    2*3=6    3*3=9
1*4=4    2*4=8    3*4=12
1*5=5    2*5=10   3*5=15
1*6=6    2*6=12   3*6=18
1*7=7    2*7=14   3*7=21
1*8=8    2*8=16   3*8=24
1*9=9    2*9=18   3*9=27
1*10=10  2*10=20  3*10=30
```



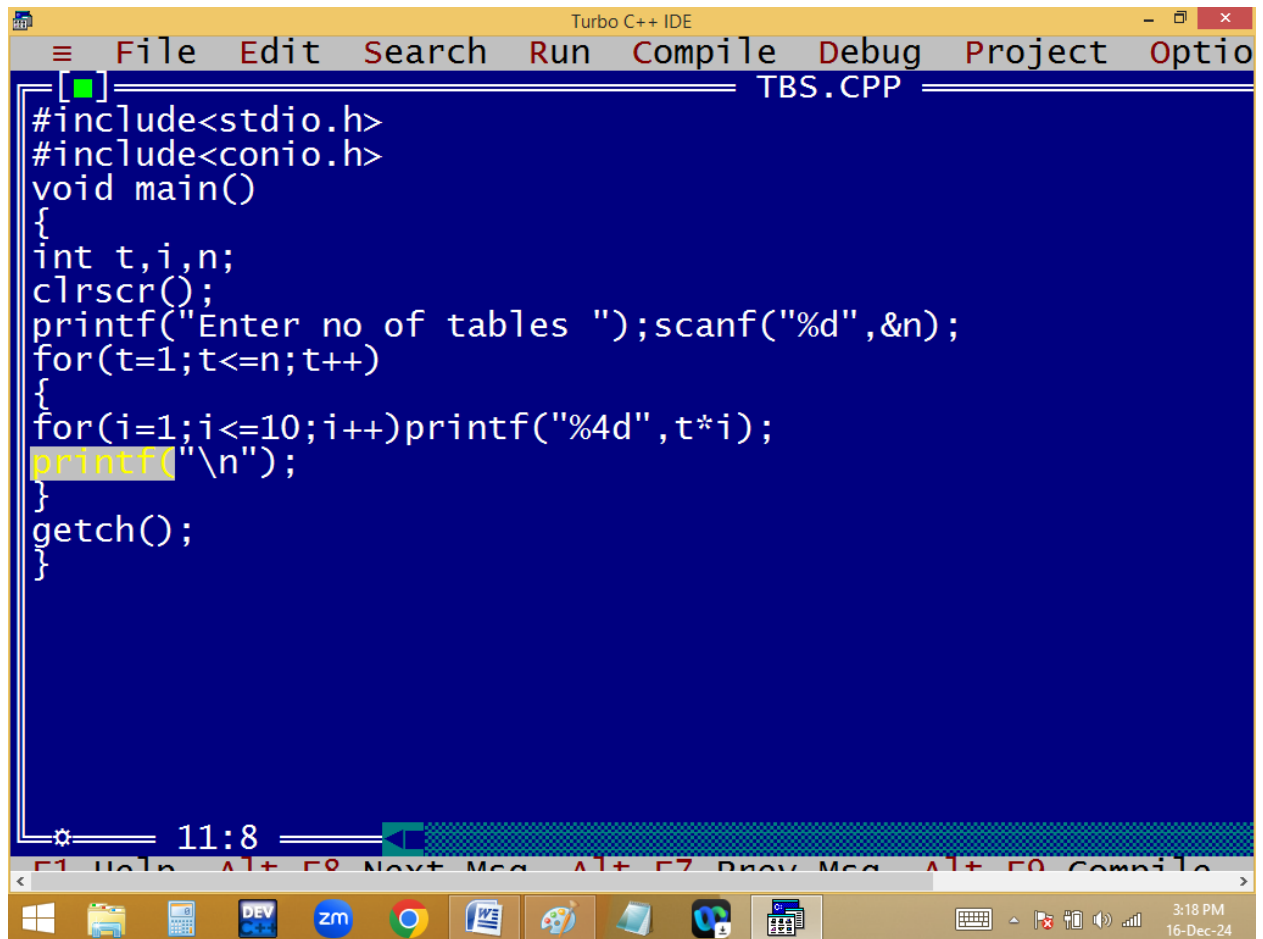
The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)
        {
            printf("%4d",t*i);
        }
        printf("\n");
    }
    getch();
}
```

The status bar at the bottom shows the time "12:31" and the date "16-Dec-24". The Windows taskbar is visible at the very bottom with various application icons.

```
Turbo C++ IDE
Enter no of tables 10
1 2 3 4 5 6 7 8 9 10
2 4 6 8 10 12 14 16 18 20
3 6 9 12 15 18 21 24 27 30
4 8 12 16 20 24 28 32 36 40
5 10 15 20 25 30 35 40 45 50
6 12 18 24 30 36 42 48 54 60
7 14 21 28 35 42 49 56 63 70
8 16 24 32 40 48 56 64 72 80
9 18 27 36 45 54 63 72 81 90
10 20 30 40 50 60 70 80 90 100
```

Windows taskbar: 3:12 PM, 16-Dec-24



The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right. The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables ");scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)printf("%4d",t*i);
        printf("\n");
    }
    getch();
}
```

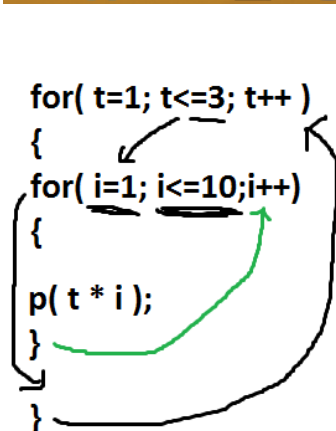
The status bar at the bottom shows the time "11:8" and a cursor icon. The Windows taskbar is visible at the very bottom with various application icons and the system clock showing "3:18 PM 16-Dec-24".



```

Turbo C++ IDE
Enter no of tables 10
1  2  3  4  5  6  7  8  9 10
2  4  6  8 10 12 14 16 18 20
3  6  9 12 15 18 21 24 27 30
4  8 12 16 20 24 28 32 36 40
5 10 15 20 25 30 35 40 45 50
6 12 18 24 30 36 42 48 54 60
7 14 21 28 35 42 49 56 63 70
8 16 24 32 40 48 56 64 72 80
9 18 27 36 45 54 63 72 81 90
10 20 30 40 50 60 70 80 90 100

```



$$\frac{t \times i}{1} = \frac{i++ \times}{2} \quad \frac{t+k}{1} = \frac{n}{3}$$

10 : 10 x 1  
 2 1 x 2  
 20 10  
 3 1 3  
 30 10 4

|         |         |
|---------|---------|
| 1*1=1   | 10*1=10 |
| 11*1=11 | 20*1=20 |
| 1*2=2   |         |

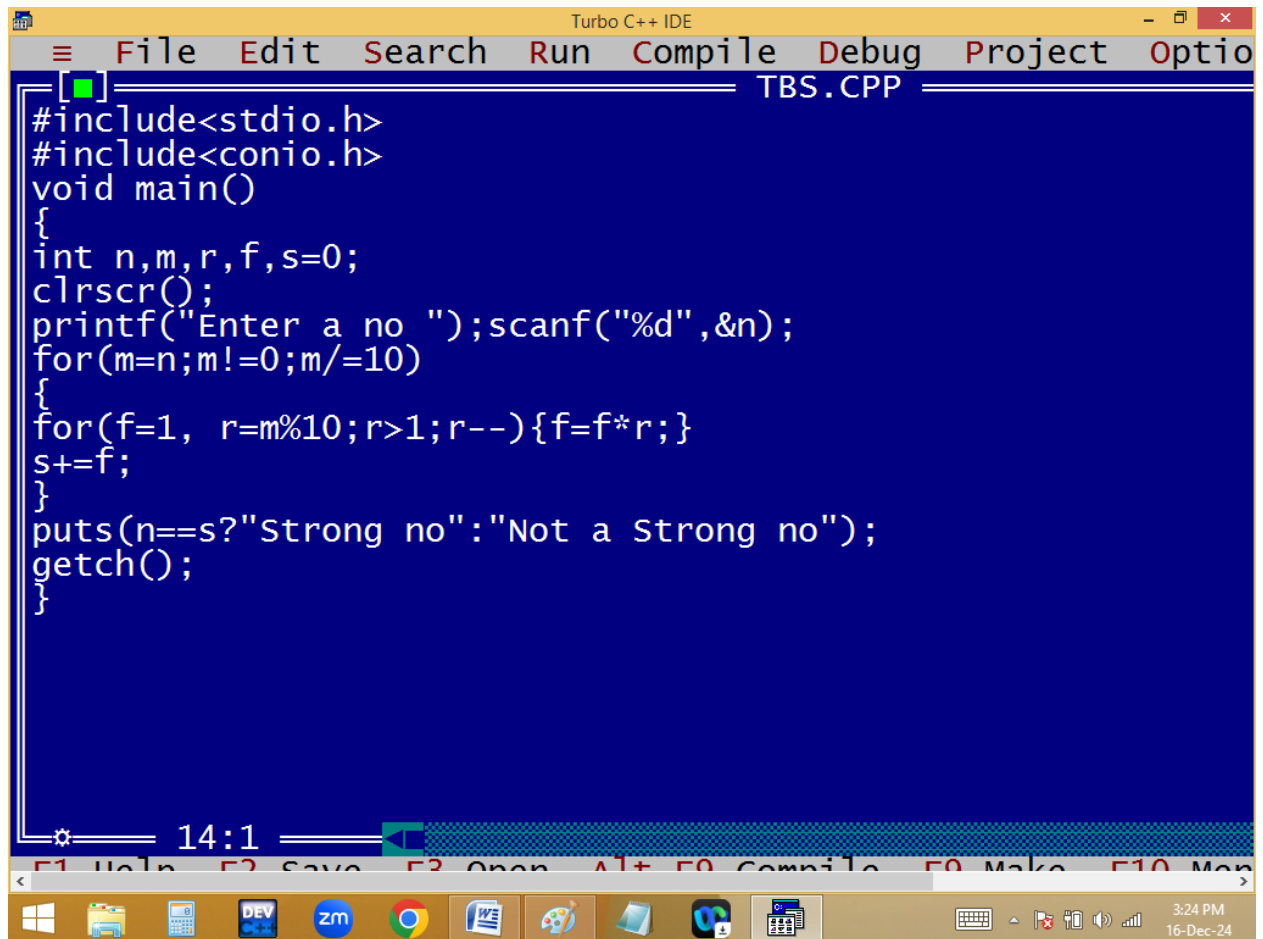
25r

Finding strong no or not?

1 factorial 1

2 factorial 2

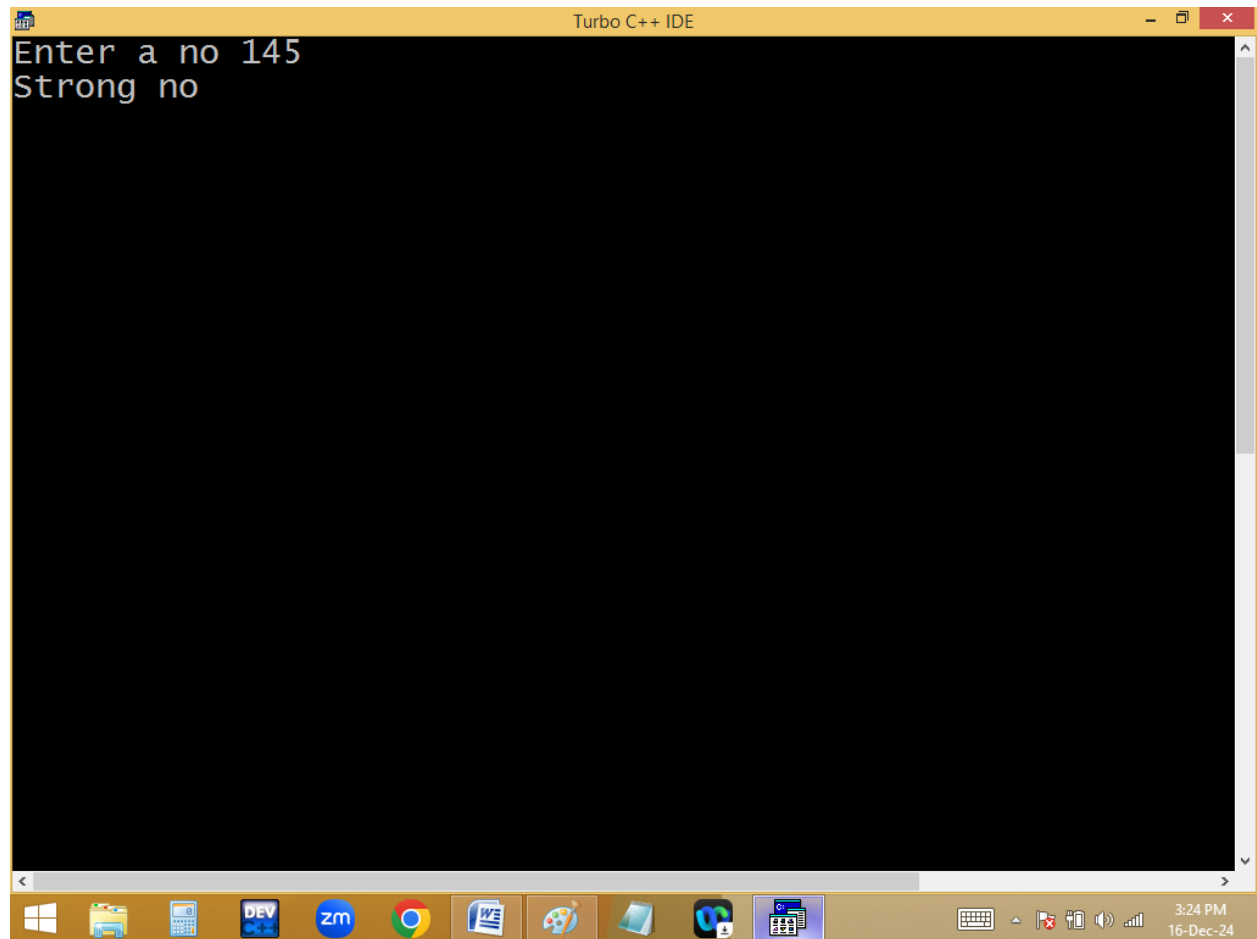
$$145 \rightarrow 1! + 4! + 5! = 1 + 24 + 120 = 145$$



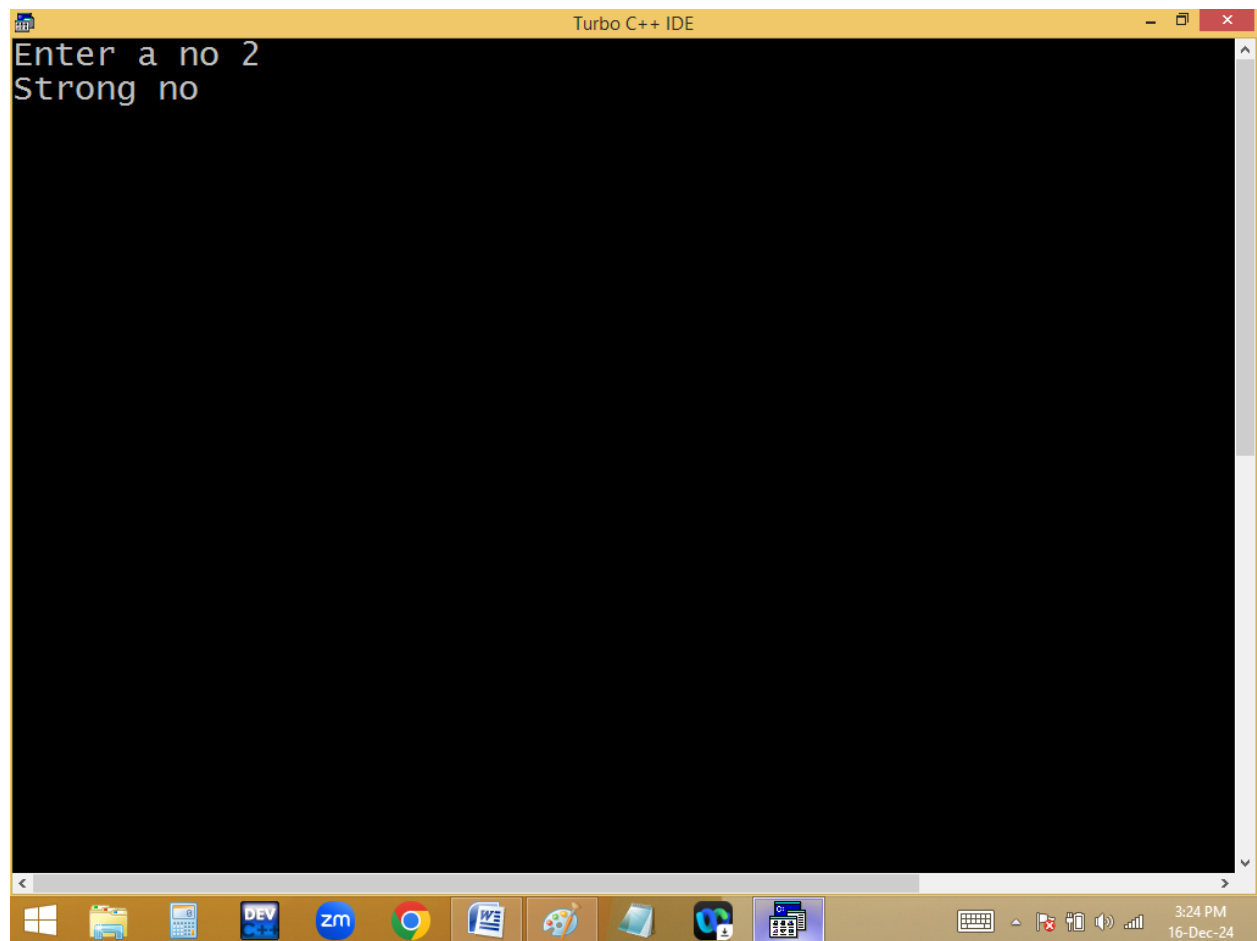
The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right of the editor area. The code in the editor is as follows:

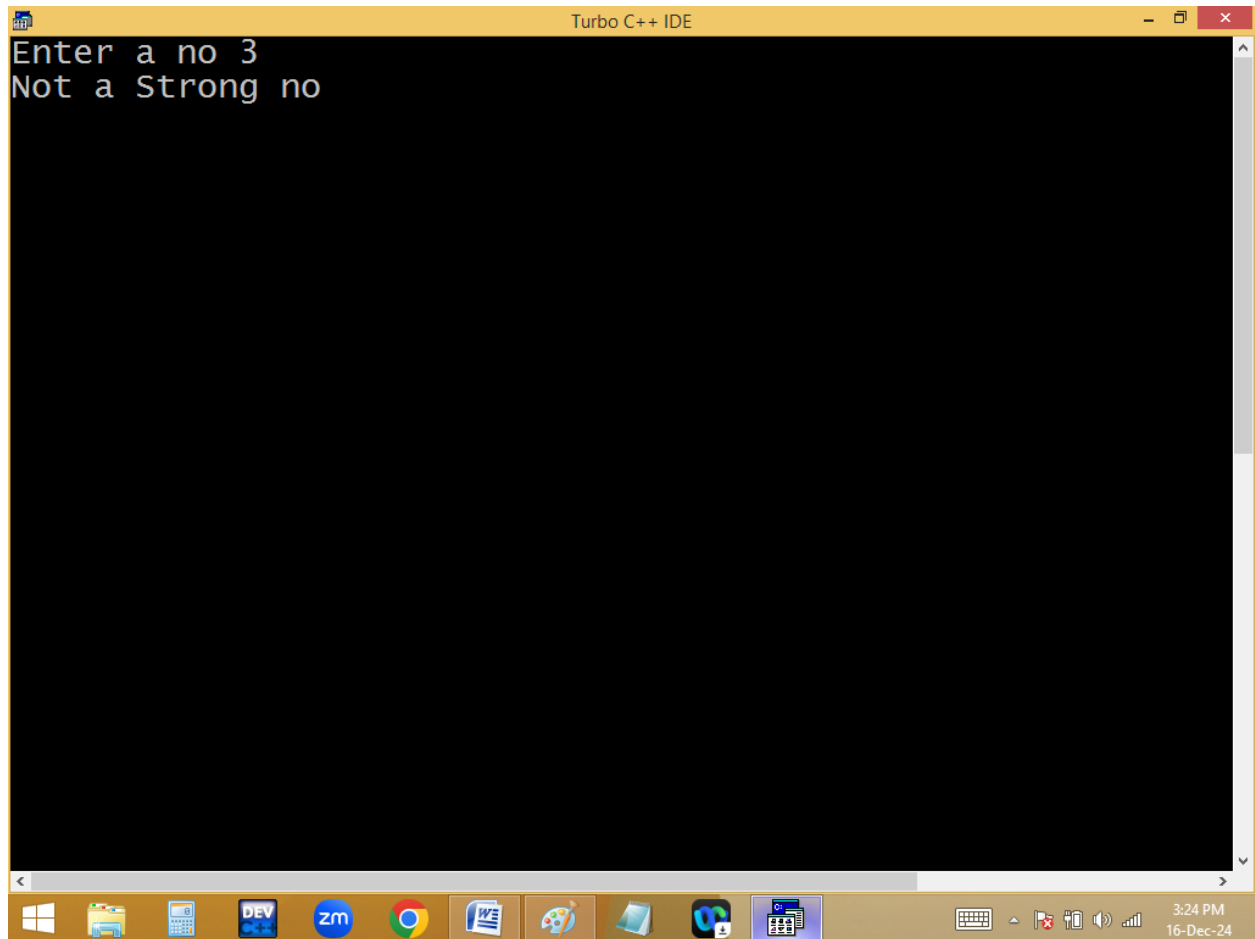
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,m,r,f,s=0;
    clrscr();
    printf("Enter a no ");scanf("%d",&n);
    for(m=n;m!=0;m/=10)
    {
        for(f=1, r=m%10;r>1;r--){f=f*r;}
        s+=f;
    }
    puts(n==s?"Strong no":"Not a Strong no");
    getch();
}
```

At the bottom of the IDE window, there is a status bar showing "14:1" and a keyboard shortcut menu with options like "F1 Help", "F2 Save", "F3 Open", "Alt+F9 Compile", "F9 Make", and "F10 Men". The Windows taskbar is visible at the very bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner indicates "3:24 PM" on "16-Dec-24".









Handwritten notes and code explaining the strong number algorithm:

**Code:**

```
for(m=n; m!=0; m/=10)
{
    for( f=1, r=m%10; r>1; r--)
    {
        f=f*r; ✓
    }
    s+=f; ✓
}
puts(n==s?"Strong":"Not");
```

**Handwritten Calculations:**

For  $n = 145$ :

$$\frac{m}{145} \div 10 = \frac{14}{5} = 2 + \frac{4}{5} = 2 + 0.8 = 2.8$$

$$14 \div 10 = 1 \text{ remainder } 4 \Rightarrow 4! = 24 + 120 = 144$$

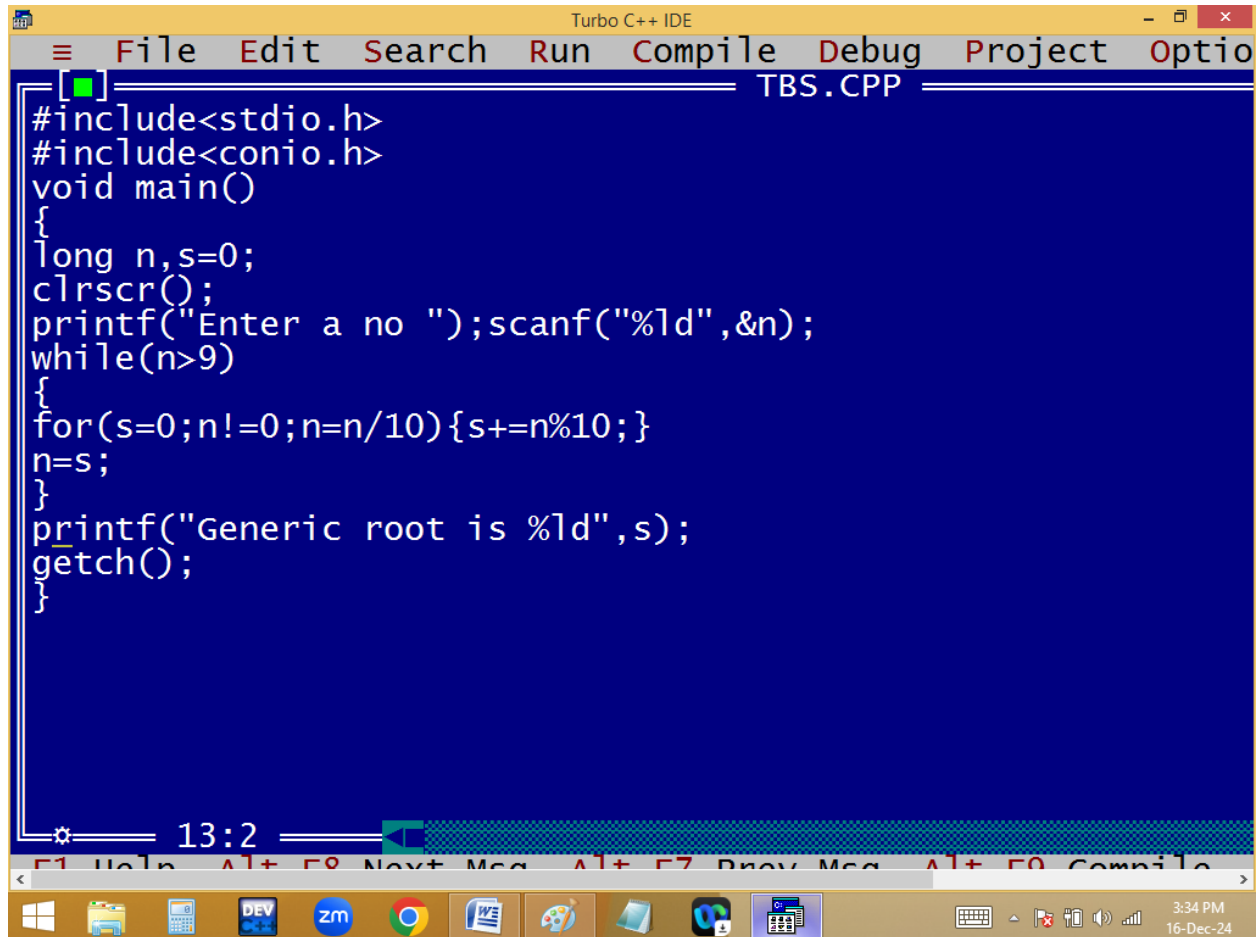
$$1 \div 10 = 0 \text{ remainder } 1 \Rightarrow 1! = 1 + 144 = 145$$

**Handwritten Multiplication:**

$$\begin{array}{r} 5432 \times \frac{f}{1} = 120 \\ 432 \times 1 = 24 \\ \hline 145 \end{array}$$

## Finding generic root of given no?

$5677 = 5+6+7+7=25 \rightarrow 2+5=7$

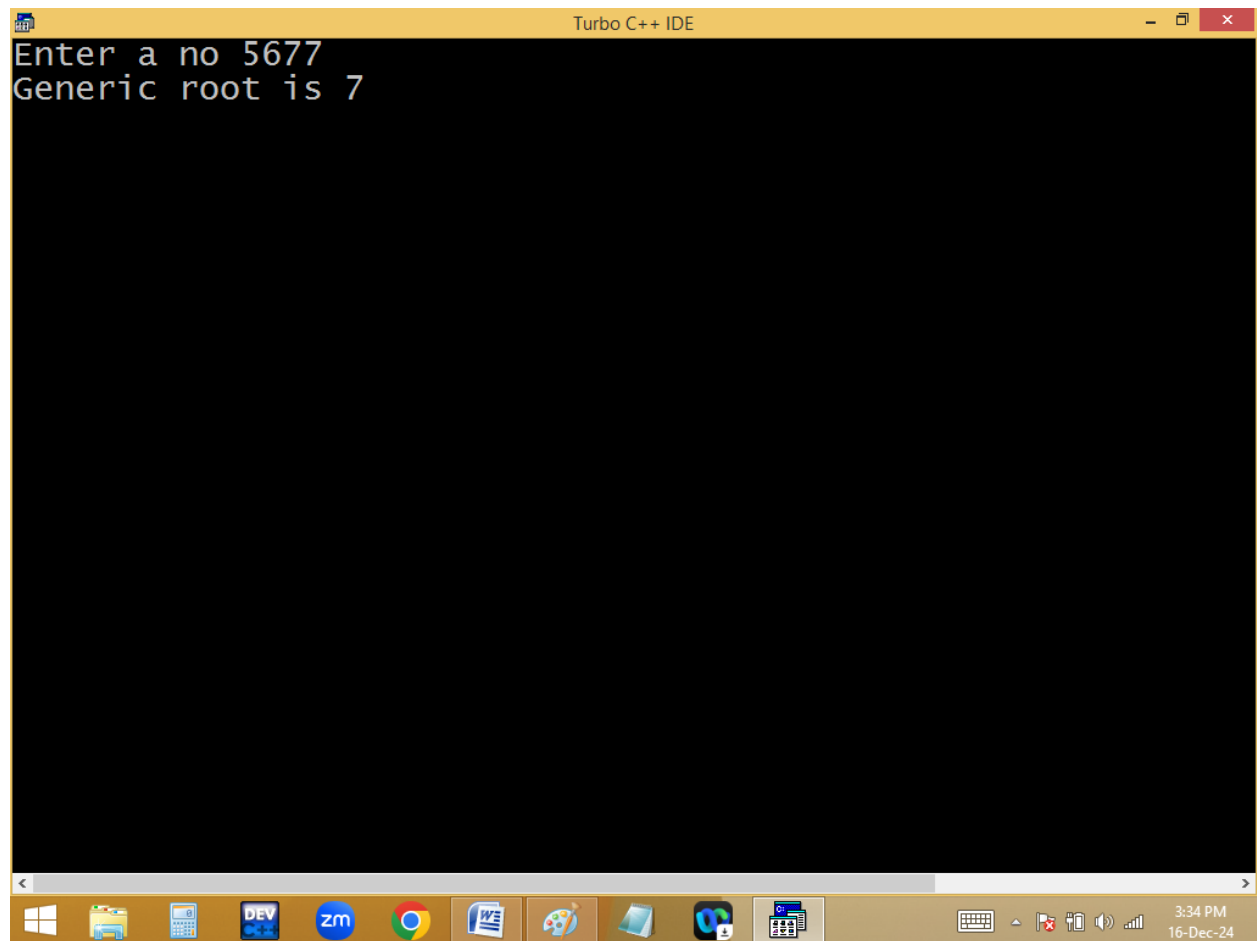


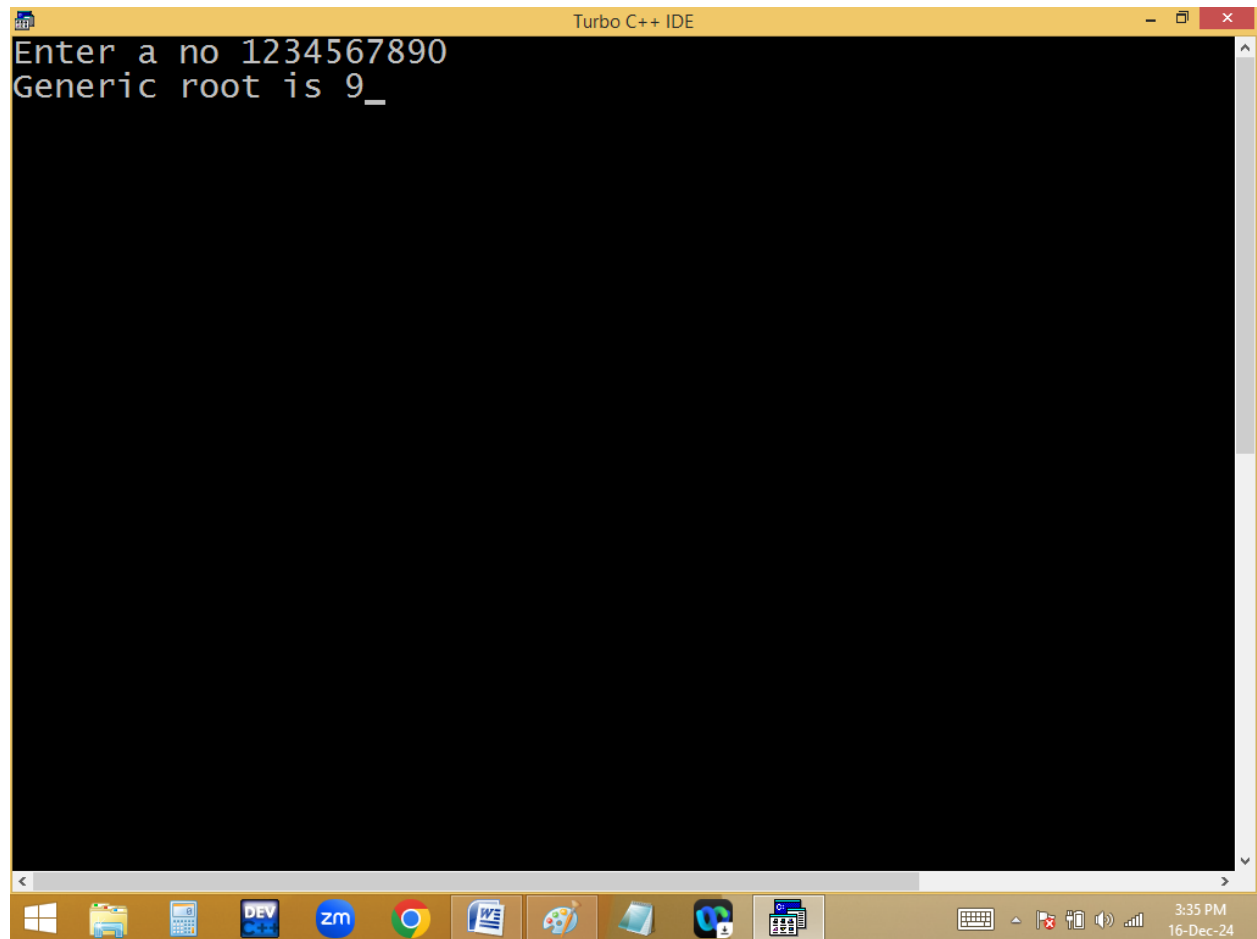
The screenshot shows the Turbo C++ IDE with a file named TBS.CPP open. The code is a C program that calculates the generic root of a given number. It includes `<stdio.h>` and `<conio.h>`. The `main` function starts by declaring `long n, s=0;`, clearing the screen with `clrscr();`, and prompting the user to enter a number with `printf("Enter a no ");scanf("%ld",&n);`. A `while` loop with condition `n>9` contains a `for` loop that repeatedly calculates the sum of digits (`s+=n%10;`) and updates `n` to the sum (`n=s;`) until the number is a single digit. Finally, it prints the generic root with `printf("Generic root is %ld",s);` and waits for a key press with `getch();`.

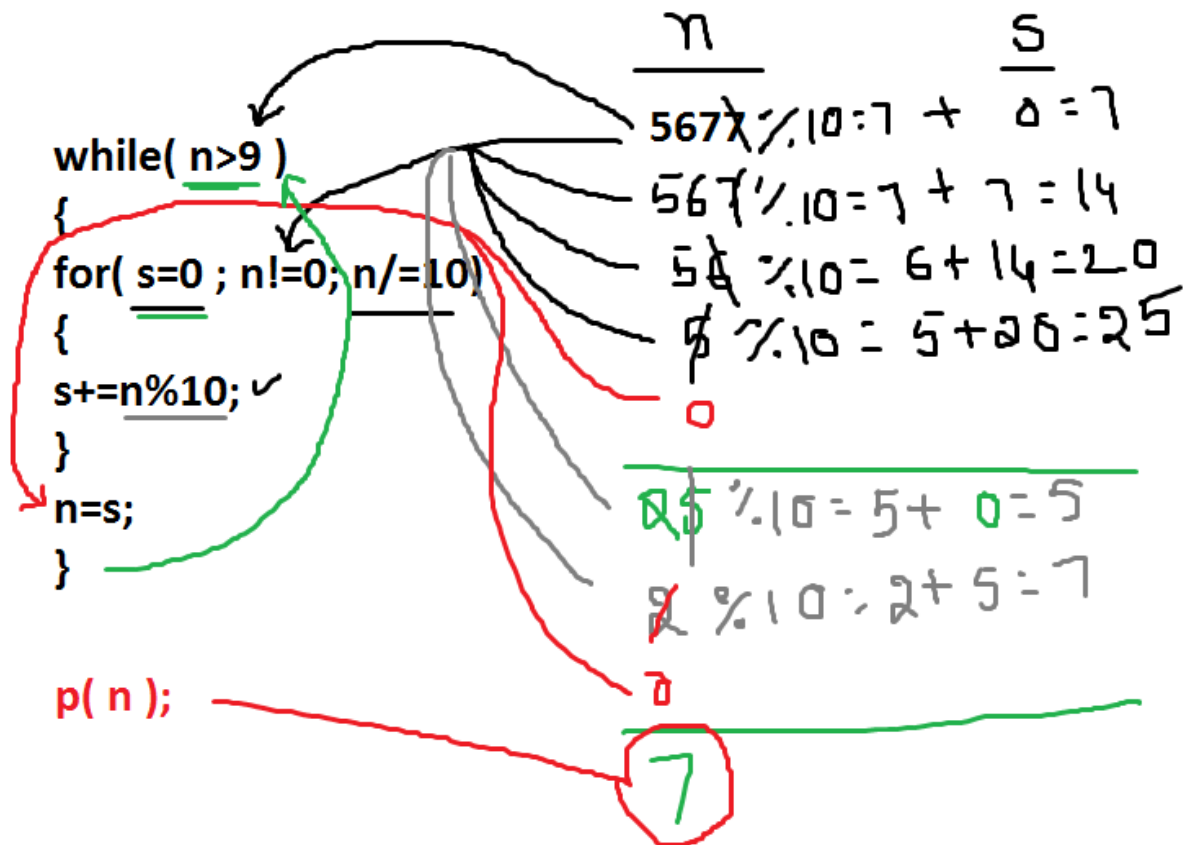
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n,s=0;
    clrscr();
    printf("Enter a no ");scanf("%ld",&n);
    while(n>9)
    {
        for(s=0;n!=0;n=n/10){s+=n%10;}
        n=s;
    }
    printf("Generic root is %ld",s);
    getch();
}
```

The IDE interface includes a menu bar with File, Edit, Search, Run, Compile, Debug, Project, and Options. The status bar at the bottom shows the time as 13:2 and the date as 16-Dec-24.



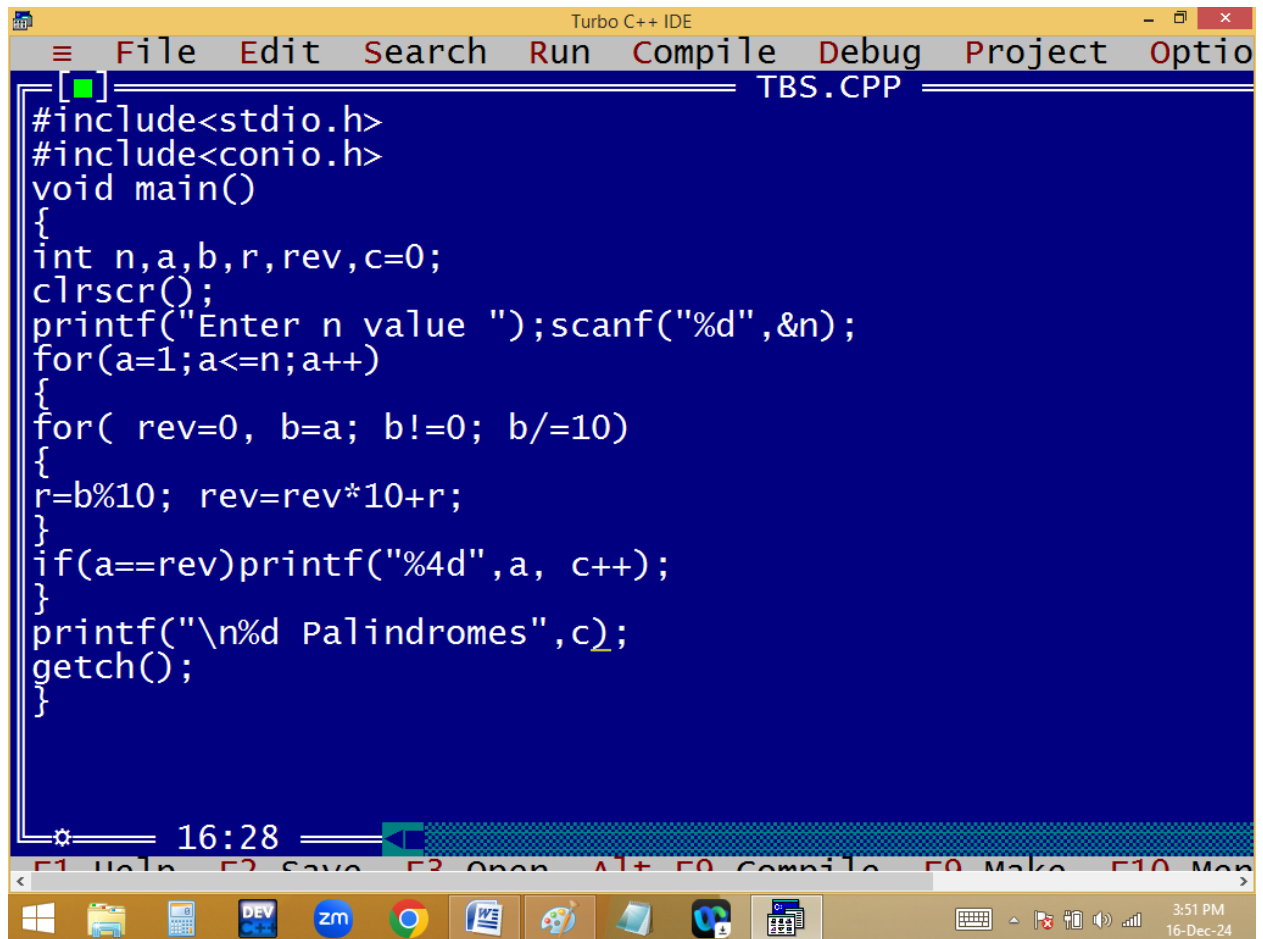






### Printing 1..n palindromes and count?

Eg: 1 to 50 palindromes are 1,2,3,4,5,6,7,8,9,11,22,33,44 →  
count 13

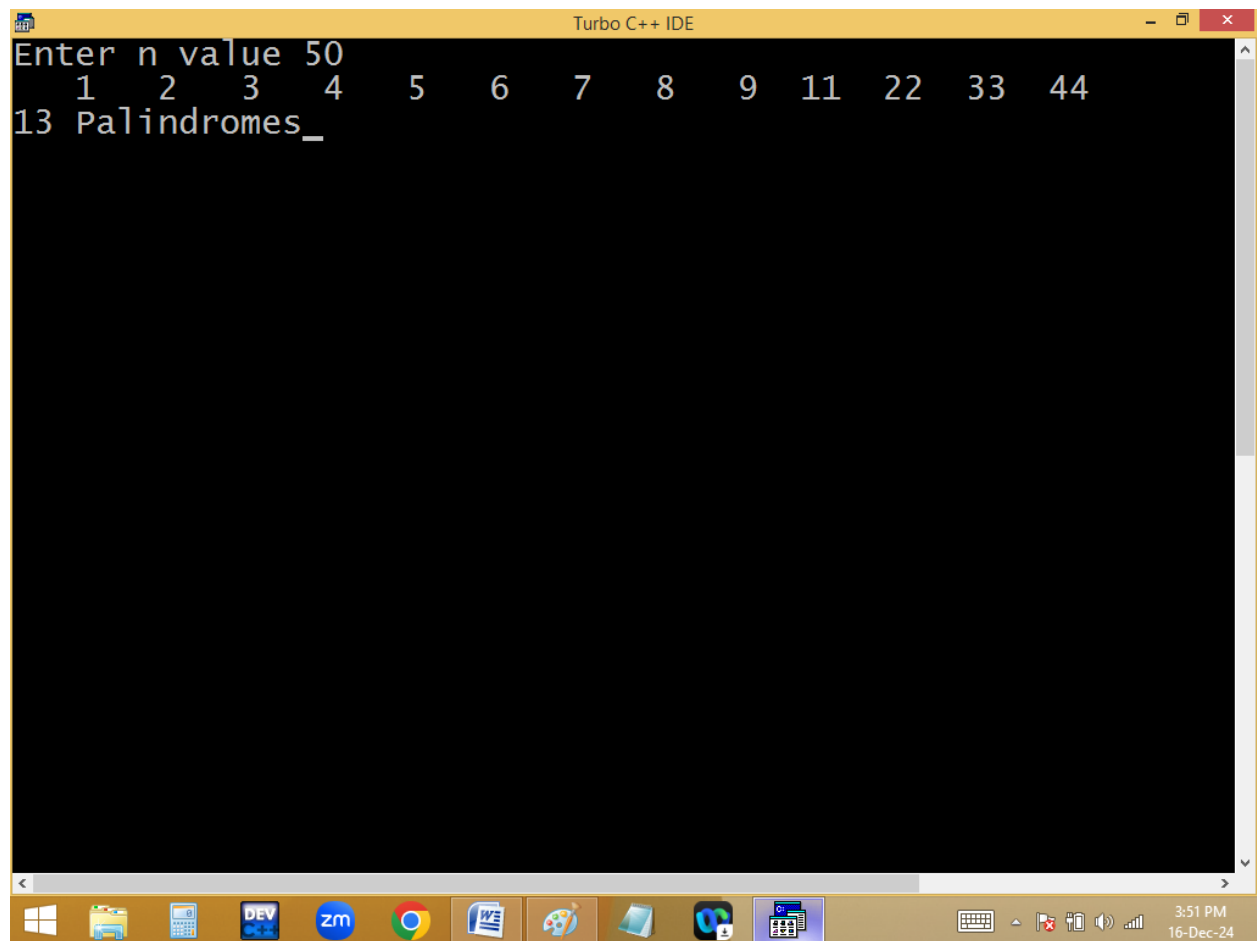


The image shows a screenshot of the Turbo C++ IDE. The title bar at the top reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "TBS.CPP" is displayed in the top right corner of the editor window. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,a,b,r,rev,c=0;
clrscr();
printf("Enter n value ");scanf("%d",&n);
for(a=1;a<=n;a++)
{
for( rev=0, b=a; b!=0; b/=10)
{
r=b%10; rev=rev*10+r;
}
if(a==rev)printf("%4d",a, c++);
}
printf("\n%d Palindromes",c);
getch();
}
```

At the bottom of the IDE window, a status bar shows the time "16:28" and a cursor icon. Below the IDE window, the Windows taskbar is visible, showing icons for various applications including a folder, calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a game controller. The system tray on the right shows the time "3:51 PM" and the date "16-Dec-24".

```
Turbo C++ IDE
Enter n value 50
1 2 3 4 5 6 7 8 9 11 22 33 44
13 Palindromes_
```





## Home work:

Finding 1..n primes and count.

Finding n to n primes and count

